#1. Introduction to Tabular ML

Instructor: **Hankook Lee** @ Efficient Learning Lab., Sungkyunkwan University

What Is Tabular Data?

- **Tabular data** is data organized into a two-dimensional structure consisting of **rows** and **columns**:
  - Each **row** (or sample, record, instance) represents a single observation or entity
  - Each **column** (or feature, attribute, variable) represents a measured property
  - A designated column may serve as the **target** (or label)

column

target

	ID	Age	Charge	Contract	Tenure	...	Churned
	1	34	$75	Monthly	6		Yes
row	2	52	$45	2-year	48		No
	3	28	$120	Monthly	2		Yes
	4	41	$60	1-year	24		No

Characteristics of Tabular Data

- **Heterogeneous Feature Types**
  - Tabular data combines **multiple feature types in a single row**
    - Each column can represent any kind of information

Type	Examples
Numerical	Age, price, temperature, number of children, …
Categorical	Country, color, category, education level, …
Binary	Yes/no, true/false, 0/1
Datetime	Timestamps, dates, …
Text (free-form)	User comments, …
Others	Image, video, …

→ Models must **handle very different statistical properties** within the input
  - This is fundamentally harder than processing homogeneous features (e.g., pixels)

Characteristics of Tabular Data

- **No Inherent Spatial or Sequential Structure**
  - In **images**, neighboring pixels are correlated → convolutions exploit this
  - In **text**, word order matters → recurrent/attention models exploit this

  - In **tabular data**, **column order is arbitrary**
    - E.g., swapping the order of "age" and "salary" should not change the meaning

  → Models must be **permutation-invariant across features**

Characteristics of Tabular Data

- **Small to Medium Dataset Sizes**
  - Modern **image** and **text datasets** can contain **billions of samples**

  - **Tabular** datasets often include **a few thousand rows**
  - The number of columns often ranges from dozens to a few thousand

  - Due to this **data scarcity**, models with strong inductive biases (e.g., tree ensembles) are often favored over data-hungry deep networks

  → Models must be **data-efficient**
  → Models must **avoid overfitting**
  → Models must **generalize** from limited and heterogeneous samples

Characteristics of Tabular Data

- **Missing Values are Common**
  - Real-world tabular data frequently contains **missing entries**
    - Sensors fail
    - Customers skip questions
    - Records come from different sources

  → Models must **handle incomplete inputs**
  → Models must be **robust to missing or unavailable features**

Characteristics of Tabular Data

- **Noisy and Imperfect Labels**
  - **Labels are often noisy** because they come from real-world processes
    - Human decisions can be inconsistent
    - Business rules may change over time
    - Labels may be collected after the input features
  - Poorly defined labels can introduce leakage
    - E.g., features measured after the outcome may reveal the target

→ Models must be **robust to noisy labels**
→ Models must **avoid leakage-prone features**
→ Evaluation must **respect temporal ordering**

Characteristics of Tabular Data

- **Class Imbalance**
  - Many tabular prediction problems have **highly imbalanced** distributions
    - Fraud detection: very few fraudulent transactions
    - Disease diagnosis: rare positive cases
    - Defect detection: few defective products
    - In extreme cases, the minority class may account for less than 1% of the data

  → Models must **avoid being biased toward the majority** class
  → Models must **detect rare but important cases**
  → Evaluation must **use appropriate metrics beyond accuracy**

Characteristics of Tabular Data

- **Domain Diversity and Limited Domain Knowledge**
  - Tabular data **appears across many domains**
    - Finance, healthcare, manufacturing, education, e-commerce
  - **Domain knowledge is often limited** or hard to encode
    - Important feature interactions may be unknown
    - Some patterns may be domain-specific or spurious

→ Models must **adapt across domains**
→ Models must **learn feature relationships from data**
→ Models must **avoid spurious shortcuts**

Characteristics of Tabular Data

- **Tabular data is everywhere**

- However, **tabular data is challenging to model** because it often has:
  - Heterogeneous feature types
  - No inherent spatial or sequential structure
  - Small to medium dataset sizes
  - Missing values, noisy and imperfect labels
  - Highly imbalanced class distributions
  - Diverse domains with limited domain knowledge

- **These characteristics make tabular learning different from other domains** such as vision and language modeling

Tasks on Tabular Data

- **Prediction** (a.k.a. Supervised Learning)
  - Given a training set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$, the goal is to **estimate an unknown target y for a new record $\mathbf{x}$**
  - Regression: $y \in \mathbb{R}$, or classification: $y \in \{1, \ldots, K\}$
  - Example: **Credit Scoring**
    - A credit score can be built from a set of credit features of borrowers

$\mathbf{x}$					y_1	y_2
ID	**OnTime**	**CreditUtil**	**Years**	**#Inquiries**	**Default?**	**Score**
1	98%	12%	18	0	No	99
2	74%	85%	3	6	Yes	50
3	91%	35%	10	2	No	90

Tasks on Tabular Data

- **Anomaly Detection**
  - **Identify rare, unusual records** that do not fit the expected pattern
  - It is usually **unsupervised**
    - We have abundant "normal" data but very few or no labeled anomalies
  - Heterogeneous features → simple distance metrics may not work
  - Example: **Credit-Card Fraud Detection**

Which record is abnormal?

TxnID	Amount	Merchant	Country	Last
1	$42.5	Coffee Shop	KR	3.0 hours
2	$38.0	Grocery	KR	2.1 hours
3	$2890.0	Electronics	RU	0.1 hours
4	$12.0	Restaurant	KR	5.4 hours

Tasks on Tabular Data

- **Clustering**
  - **Group similar records together without labels**
  - The goal is to **uncover hidden structure**
    - E.g., customer groups, operational regimes, patient subtypes
  - Common algorithms: k-means, GMMs, DBSCAN, hierarchical clustering
  - Example: **Customer Segmentation for Marketing**
    - After clustering, three customer groups might be discovered

Customer	Avg Purchase	Visits	Online Ratio	Preferred Category
1	$180.0	8	95%	Electronics
2	$45.0	12	10%	Groceries
3	$320.0	2	80%	Luxury
4	$52.0	14	5%	Groceries

Tasks on Tabular Data

- **Table Question Answering**
  - **A natural-language question and a table → return an answer**
    - This may require lookup, filtering, aggregation, or multi-step reasoning **over the table**
  - This requires both natural-language understanding and tabular reasoning
    - Modern LLMs have made it feasible
  - Example: **Internal Sales Analytics Assistant**

Order ID	Region	Quarter	Product	Revenue
1	Asia	Q1	A	1200
2	Asia	Q2	B	2400
3	Europe	Q1	A	800
4	Asia	Q3	A	1800
5	Europe	Q3	B	1100

(Q) What was total revenue in Asia in Q3?

↓

(A) Asia's Q3 revenue was 1,800

Tasks on Tabular Data

- **Synthetic Data Generation**
  - **Generate fake-but-realistic tabular records** that preserve the statistical properties of the original data while protecting individuals
  - Uses:
    - **Sharing data** across organizations **under privacy constraints**
    - **Augmenting small datasets** to improve downstream models
  - Example: **Privacy-Preserving Healthcare Data Sharing**
    - Want to share a patient dataset for research, but cannot legally release real records

Patient ID	Age	BMI	BPSystolic	Diabetes?
1	62	28.4	148	Yes
2	35	22.1	120	No
3	71	31.5	160	Yes

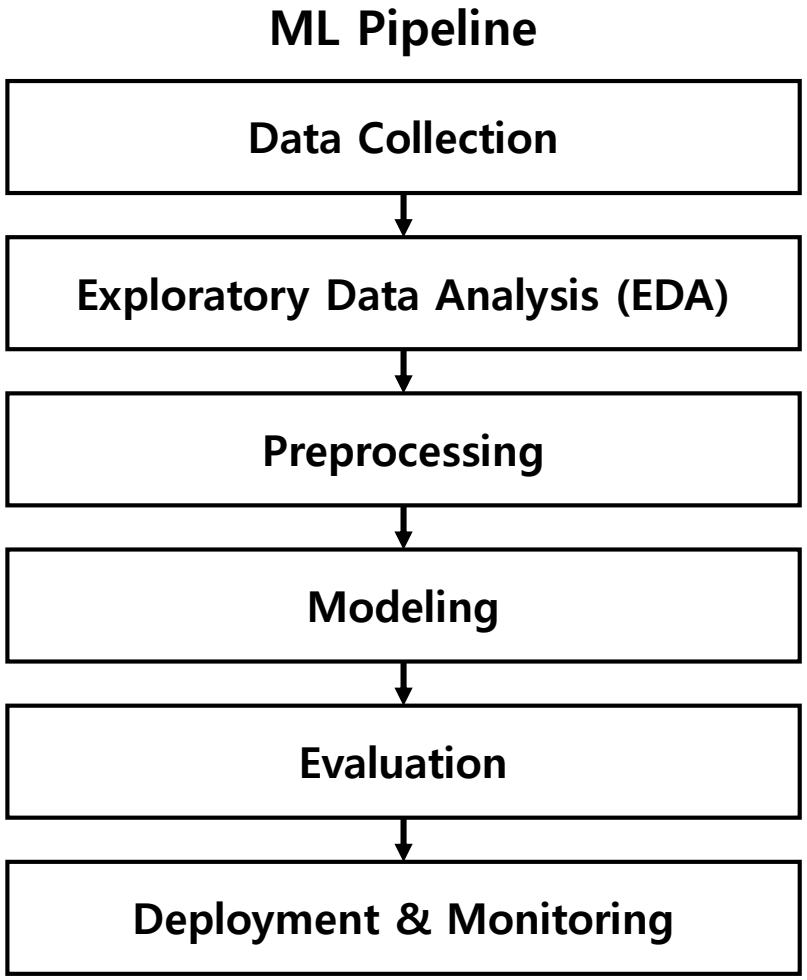
Tasks on Tabular Data

- **Tabular data is everywhere**
  - Most of the data organizations collect, and most of the decisions they make from data, take a tabular form

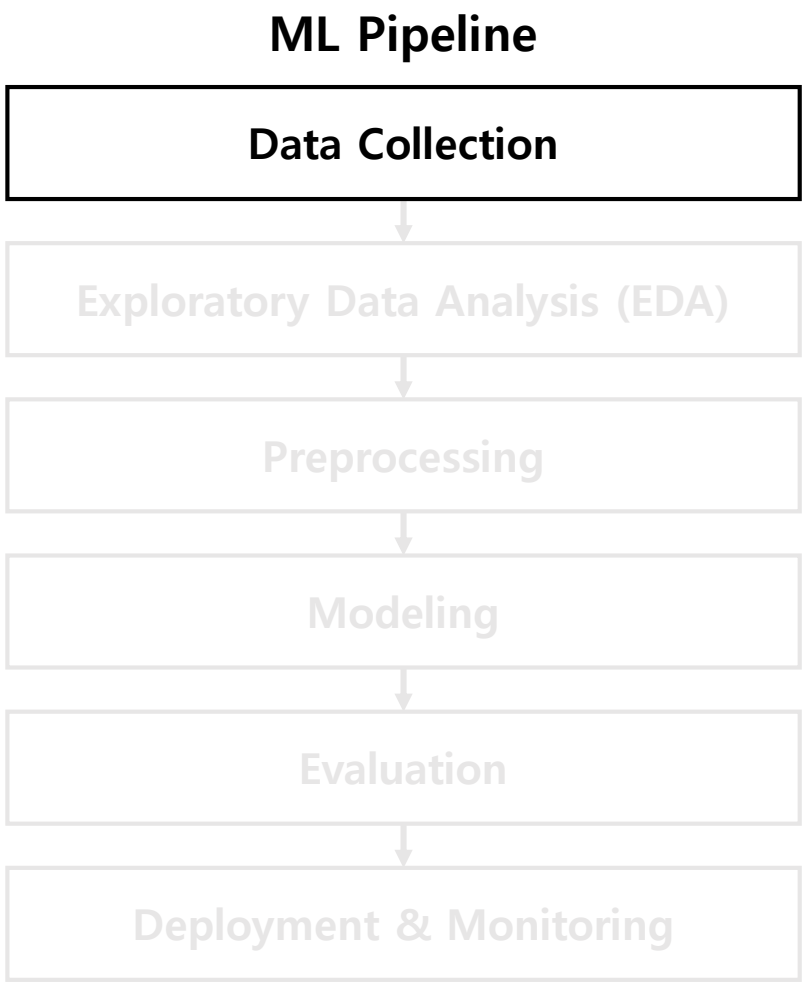
- **Many real-world problems can be expressed as tabular tasks**
  - A wide range of applications across different domains and goals can be cast in the row-and-column form we have seen

(Q) How do we actually apply machine learning to a real tabular problem, from raw data to a deployed, trustworthy model?

Machine Learning Pipeline

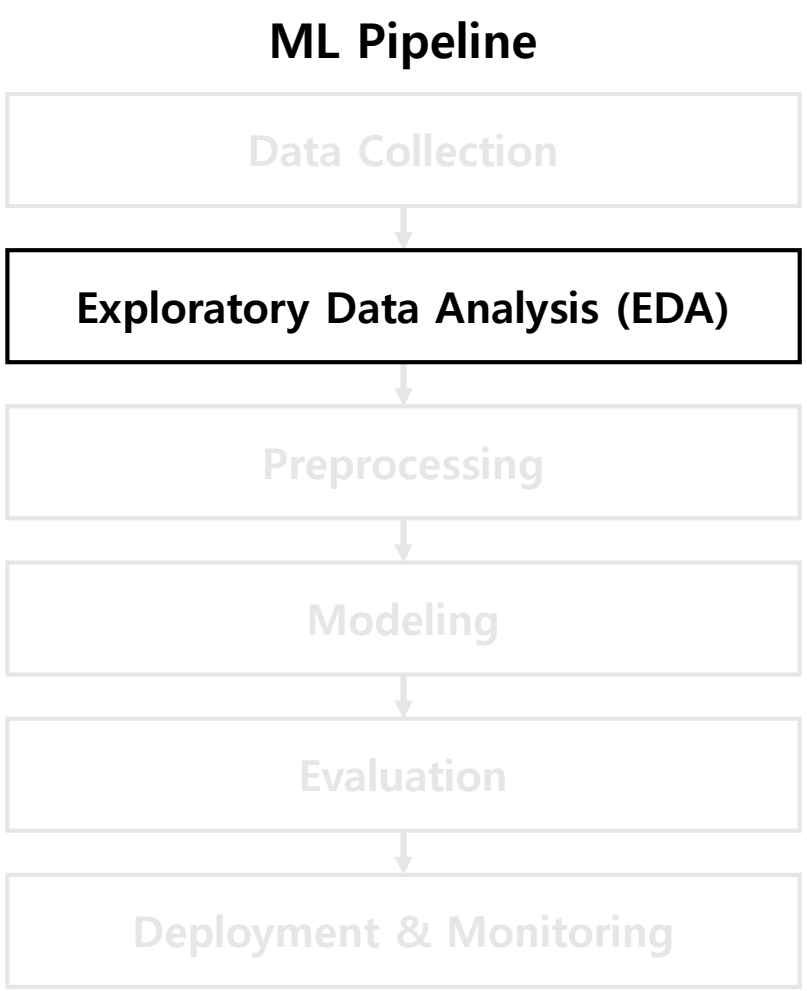


Machine Learning Pipeline



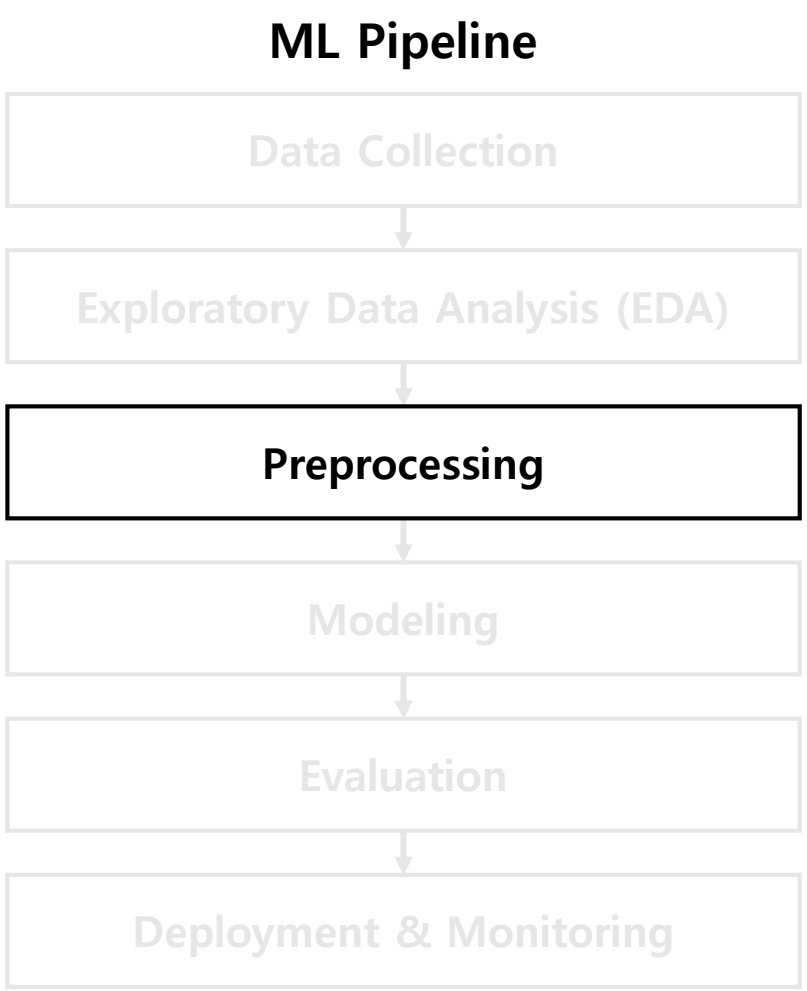
- **Data Collection**
  - **Collecting relevant data from reliable sources** is the first step in an ML pipeline
  - **Combine information from multiple tables**
    - E.g., customers ↔ orders ↔ products
  - **Incorporate useful external data sources**
    - E.g., timestamps ↔ weather information
  - Ensure **reproducibility**
    - Clearly document how, when, and from where the data was collected

Machine Learning Pipeline



- **Exploratory Data Analysis (EDA)**
  - Before modeling, **understand the data**:
    - Summary statistics (mean, median, quartiles, std)
    - Missing-value patterns (which features? how many?)
    - Distribution plots (histograms, KDE)
    - Correlation analysis between features and with the target
    - Potential outliers and class imbalance
  - **EDA helps you build intuition about the data** and identify potential issues before modeling

Machine Learning Pipeline

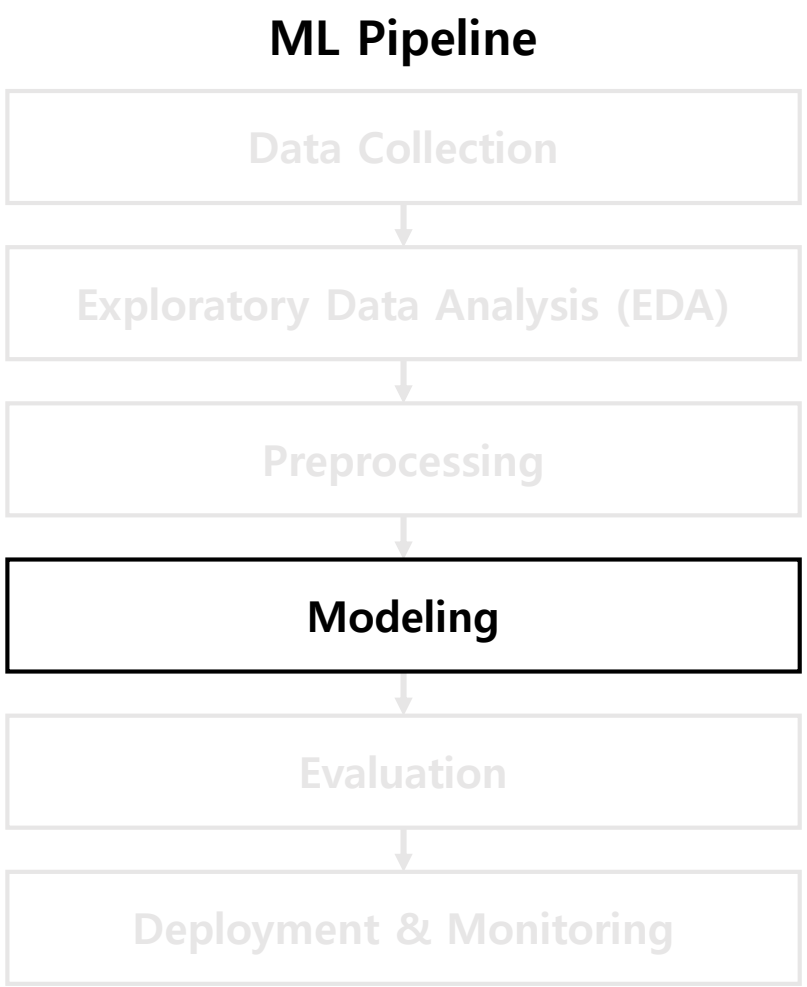


- **Preprocessing**
  - **Transform raw data into a form that a model can understand/learn from**
  - **Preprocessing is a critical step** in building a reliable ML pipeline
    - Even a powerful model may perform poorly if the input data is not properly prepared
  - **Avoid data leakage**
    - Fit preprocessing steps using the training data only
    - Apply the fitted transformations to validation/test data

```
# Wrong
X = scaler.fit_transform(X)
X_train, X_test = split(X, ...)
```

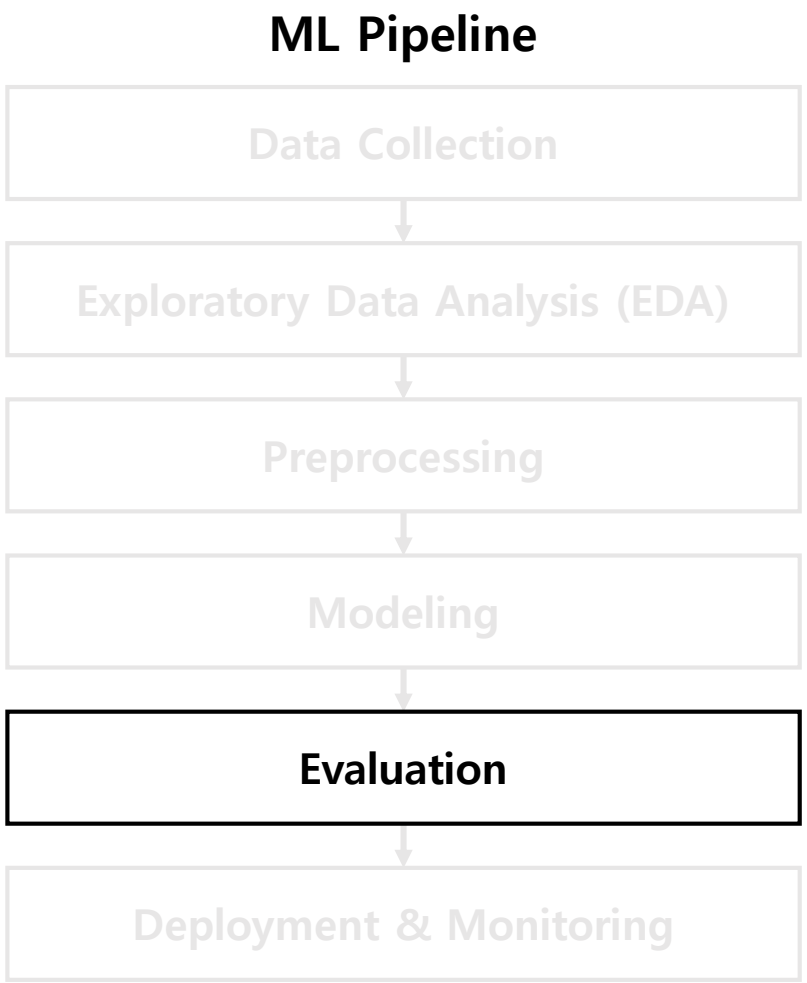
```
# Correct
X_train, X_test = split(X, ...)
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Machine Learning Pipeline



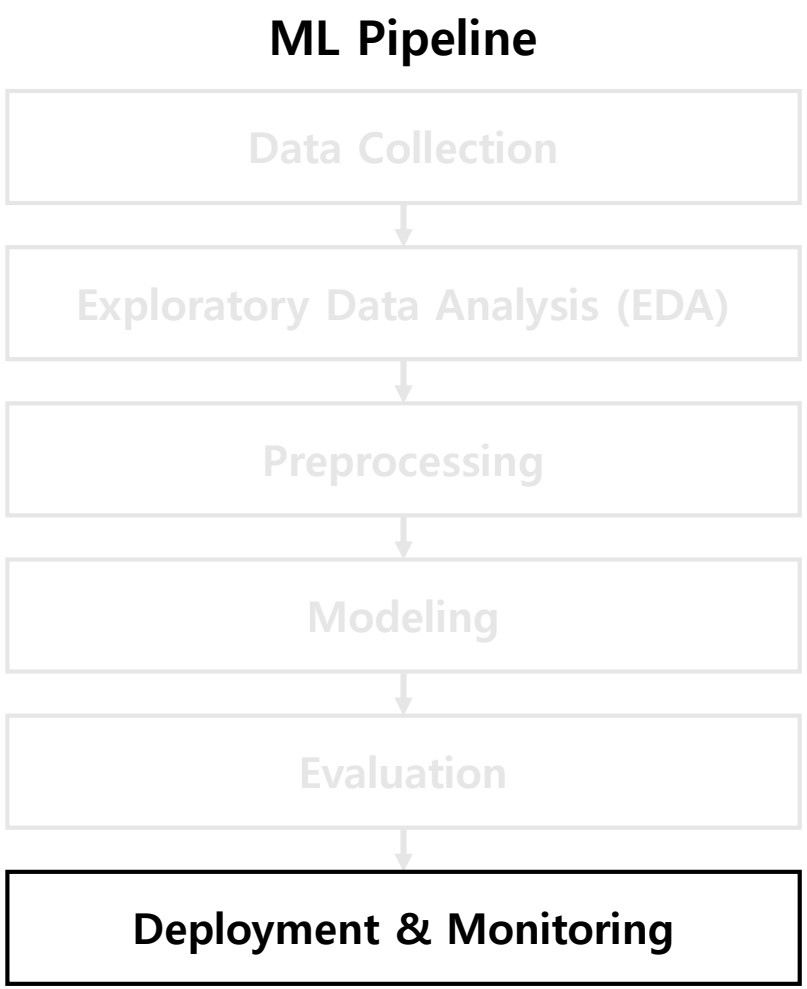
- **Modeling**
  - **Choose and train a model**
  - Linear/logistic regression (interpretable baselines)
  - Tree-based ensembles
    - E.g., XGBoost, LightGBM, CatBoost
  - Deep neural networks and foundation models
    - LLMs & TabPFNs
  - Each will be studied in later lectures

Machine Learning Pipeline



- **Evaluation**
  - Use a **held-out** test set for final evaluation
  - Use **cross-validation** for more reliable assessment
    - Especially useful when the available dataset is limited
  - **Choose an appropriate metric** for the task
    - Imbalanced classification → accuracy may be misleading
    - Regression → MAE or RMSE
  - Evaluation should reflect the real-world goal

Machine Learning Pipeline



- **Deployment**
  - Consider real-world scenarios: batch vs online inference
  - Ensure consistency btw training & deployment
    - E.g., the same preprocessing steps must be applied
- **Monitoring**
  - Track data drift, concept drift, performance degradation, and fairness metrics
  - Retrain or update the model when needed

Tabular Data Preprocessing

- Due to the heterogeneity of features,
  data preprocessing plays a crucial role in tabular ML

- You MUST consider the following before applying ML:
  - Handling missing values
  - Encoding categorical variables
  - Numerical feature transformations
  - Feature engineering

Tabular Data Preprocessing

- **Handling Missing Values**
  - **Deletion**: Remove rows or columns with missing values → lose information
  - **Constant fill**: Replace with "Unknown" → preserves missingness signals
  - **Statistical imputation**: Replace with the mean, median, or mode
  - **Model-based imputation**: Predict missing values using other features
    - E.g., use k-NN for imputation
  - **Native support**: Some tree-based models handle missing values directly

Tabular Data Preprocessing

- **Encoding Categorical Variables**
  - ML models require numerical input → categorical features must be encoded

Method	Basic Idea
One-Hot Encoding	Create one binary feature for each category
Ordinal Encoding	Map categories to ordered integers
Target Encoding	Replace each category with its average target value
Embedding Encoding	Learn a vector representation for each category

  - **One-hot encoding** is simple and does not impose an artificial order
    - For high-cardinality features, it creates too many columns → require compact encodings
  - **Ordinal encoding** is sometimes useful when categories have a natural order
    - E.g., T-shirt size: small < medium < large < x-large

Tabular Data Preprocessing

- **Numerical Feature Transformations**
  - **Scaling**: Many models are sensitive to feature scale
    - **Standardization**: $x' = (x - \mu)/\sigma$ – zero-mean, unit variance
    - **Min-max scaling**: $x' = (x - \min)/(\max - \min)$ – maps to $[0,1]$

    - Some models are scale-invariant (e.g., tree-based models)
    - Scaled values may be less directly interpretable

  - **Distribution transformations**
    - **Log transform**: useful for right-skewed variables (e.g., income, prices, counts)
    - **Quantile transform**: map values to their ranks in the distribution – robust to outliers

  - **Binning (Discretization)**
    - This may help linear models capture non-linearities

Tabular Data Preprocessing

- **Feature Engineering**
  - **Create informative features** from raw data using domain knowledge
  - Well-designed features can make important patterns easier for modeling

Pattern	Example
Aggregations	Average purchase amount over the last 30 days
Ratios	Debt-to-income ratio
Time differences	Days since last login
Date decomposition	Month, day of week, or holiday indicator
Interactions	Price * Quantity
Domain-specific features	BMI in healthcare, financial indicators in finance

  - Feature engineering can sometimes matter more than model choices

Evaluation for Tabular ML

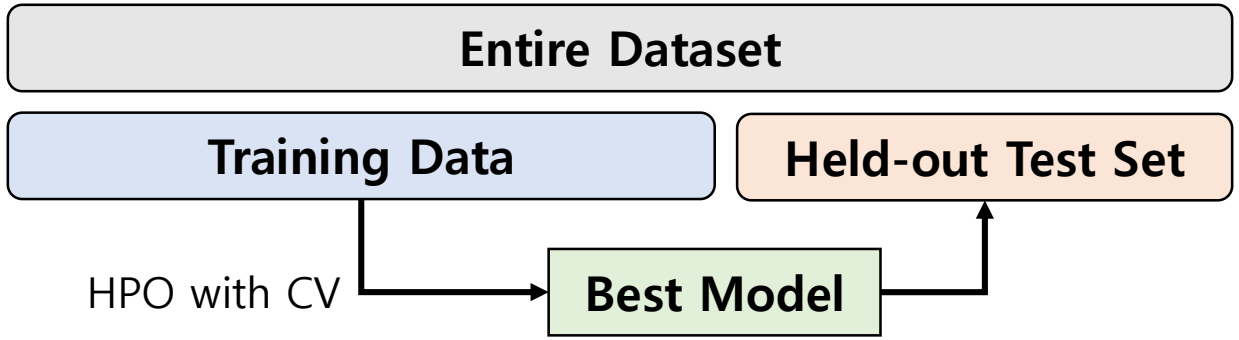
- Tabular ML can be **highly sensitive to hyperparameter choices**
  - Tabular datasets are often relatively small and exhibit heterogeneous, irregular data patterns

- **Evaluation is essential for reliable model selection**
  - Cross-validation and hyperparameter optimization
  - Choosing the right evaluation metrics

  - A strong evaluation protocol is as important as choosing a strong model

Evaluation for Tabular ML

- Cross-validation (CV) and hyperparameter optimization (HPO)
  - Use a **held-out test set** for final evaluation
  - Use **cross-validation** to obtain a more reliable validation score
  - **Tune hyperparameters** based on cross-validation performance



- Common Pitfalls
  - **Train-test contamination**: using test data during training & preprocessing
  - **Overfitting to one validation set**: tuning thousands of hyperparameter combinations on a single split

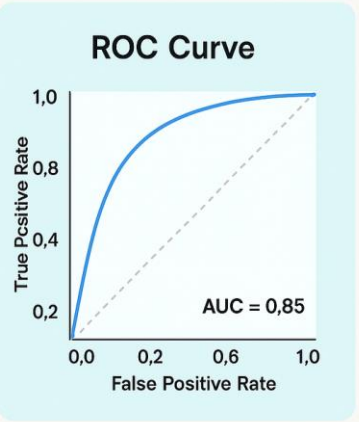
Evaluation for Tabular ML

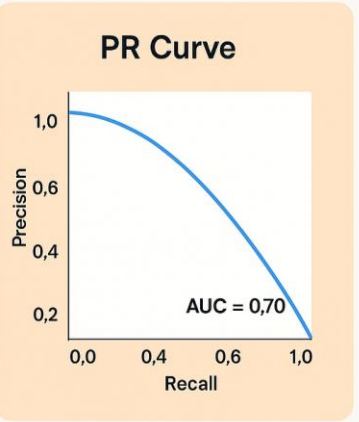
- Evaluation metrics
  - **Regression**: MAE, MSE, R^2 score $= 1 - (\sum_i (y_i - \hat{y}_i)^2)/(\sum_i (y_i - \bar{y})^2)$
  - **Binary classification**: Accuracy, Precision, Recall, F1 Score

	Pred. Positive	Pred. Negative
Real. Positive	TP	FN
Real. Negative	FP	TN

Accuracy = (TP + TN) / total
Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1 Score = harmonic mean of precision and recall

  - **Threshold-independent metrics**: AUC-ROC, AUC-PR
    - AUC-PR is more informative than AUC-ROC for highly imbalanced data





Evaluation for Tabular ML

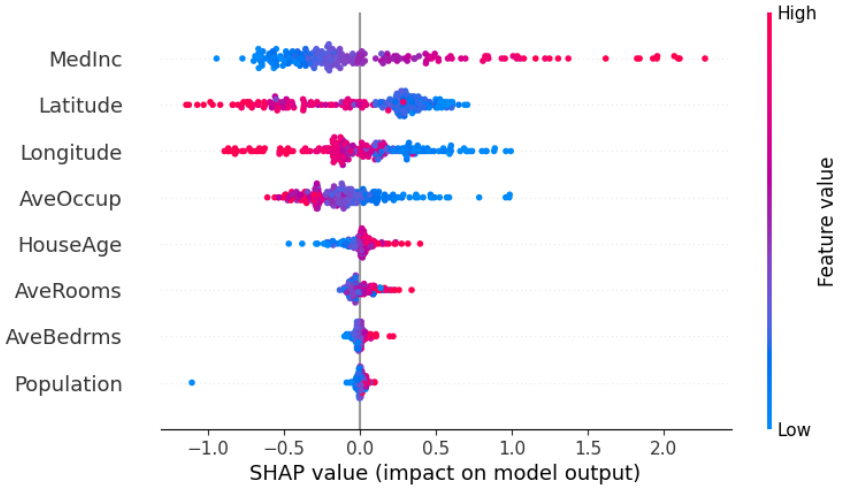
- **Choose the right metric** for your problem
  - **Fraud detection**: very rare positives → use PR-AUC, recall at fixed precision
  - **Medical screening**: missing a disease is catastrophic → optimize recall
  - **Spam filtering**: false positives are costly → optimize precision

  - The best metric is the one that reflects the real-world cost of errors

Interpretability

- We often need to **understand why a model makes a prediction**
  - High predictive performance alone is not enough
  - **Globally**, which features matter to the model overall?
  - **Locally**, why this prediction for this sample?

- **Key techniques**
  - **Permutation Importance**: Performance drop after shuffling a feature
  - **SHAP**: Feature contributions to predictions
    - Especially efficient for tree-based models



Tools & Datasets

- **Tools**
  - NumPy, Pandas, scikit-learn, …
  - AutoML libraries (e.g., Optuna), Deep Learning libraries (e.g., pytorch), …

- **Public Datasets and Benchmarks**
  - Kaggle
  - UCI Machine Learning Repository
  - OpenML
  - AI Hub (aihub.or.kr)
  - …

Summary & Lecture Roadmap

- **Tabular data is everywhere**
- Yet, **learning from tabular data is still challenging**
  - Heterogeneous features, missing values, limited data, imbalanced labels, and domain-specific patterns

- In the following lectures, we will cover:
  - Classical ML for Tabular Data
  - Deep Architectures for Tabular Data
  - Tabular Representation Learning
  - LLMs with Tabular Data
  - A New Paradigm: TabPFN

Thank You for Your Attention!